INCH-POUND

MIL-R-49276A(CR)

16 August 1988

SUPERSEDING
MIL-R-49276(CR)
15 FEB 1985

MILITARY SPECIFICATION
RADIO REPEATER SET
AN/TRC-138A

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the performance and test requirements and qualifications of the air or vehicular transportable Radio Repeater Set, AN/TRC-138A. This assemblage provides facilities for either a repeater capability to extend the range of a radio system, or a terminal capability with provisions for terminating system at the radio park.

1.2 Description. Radio Repeater Set Assemblage AN/TRC-138A, hereinafter called the shelter assemblage, includes the Shelter, Electrical Equipment S-667/TRC-138A and all operational components comprising the shelter assemblage. The S-667/TRC-138A hereinafter called the shelter facility, includes a modified S-280C/G shelter, and all racks, brackets, shelves, and tie-downs, patch panel, electrical components and wiring to enable installation and operation of the shelter assemblage. (See 6.1 and 6.5)

2. APPLICABLE DOCUMENTS.

2.1 Government Documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, assitions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US ARMY Communication-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5820

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SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies, Electrical
MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys and Titanium, Spot and Seam
MIL-T-7928	Terminals, Lug and Splice, Crimp Style Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-W-45205	Welding, Gas, Metal-Arc and Gas, Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor
MIL-S-55507	Shelter, Electrical Equipment (with or without equipment), Packaging of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics Measurement of
MIL-STD-810	Environmental Test Methods
MIL-STD-1235	Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Order Of Precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supercede applicable laws and regualations unless a specific exemption has been obtained.

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3. REQUIREMENTS.

3.1 First Article. When specified in the contract, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. The shelter facility shall be constructed in accordance with drawings as listed on DL-A3011096 and the requirements specified herein, unless otherwise specified in the contract. (See 3.22)

3.3 Electrical Installation. (See 4.6)

3.3.1 Power System Wiring and Components. The completed in-place power distribution system shall meet the following requirements:

3.3.1.1 Continuity of Conductors. Continuity shall exist between each pin of the power entrance connectors and the respective circuit terminations and shall be less than three (3) ohms. (See 4.6.1 and 4.6.1.1)

3.3.1.2 Dielectric Breakdown and Insulation Resistance. All power wires cables, and components shall withstand without flashover or insulation breakdown, a potential of 500 volts DC between each of the ungrounded power line conductors and shelter facility ground applied continuously for not less than thirty (30) seconds. The insulation resistance shall not be less than 100 megohms. (See 4.6.1 and 4.6.1.2)

3.3.2 Power Distribution and Operational Functions. Main power branch distribution and circuit switching functions thereof shall meet the operational requirements when tested in accordance with 4.6.2.

3.3.3 Signal system Wiring and Components. The completed in-place signal system installation shall meet the following requirements:

3.3.3.1 Continuity. Continuity shall exist between end terminations of all signal wires and shall be less than three (3) ohms, unless otherwise specified herein. (See 4.6.3 and 4.6.3.1)

3.3.3.2 Dielectric Breakdown and Insulation Resistance. All signal wires, cables and components shall withstand without flashover or insulation breakdown, a potential of 500 volts DC applied continuously for not less than five (5) seconds. The insulation resistance shall not be less than 100 megohms. (See 4.6.3.2)

3.3.4 Video System and Radio Frequency (RF) Systems Wiring and Components. Installation of video and RF cable assemblies, and connecting components (including plugs, connectors, adapters, receptacles, and lightning protectors) shall be in accordance with the drawings unless otherwise specified. The completed in-place video and RF system installation shall meet the following requirements:

3.3.4.1 Continuity. The point-to-point continuity of all video cable assemblies shall not exceed 0.5 ohms. (See 4.6.4 and 4.6.4.1)

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3.3.4.2 Dielectric Breakdown and insulation Resistance. The insulation of all video cables, components, and lightning protectors shall withstand the following potentials with the insulation resistance not less than 100 megohms. (See 4.6.4.2)

a. 1500 volts DC applied continuously for not less than thirty (30) seconds to all video cable assemblies terminated at the video entrance box with lightning protector element removed.

b. 790 volts DC applied continuously for not less than thirty (30) seconds to all video cable assemblies terminated at the video entrance box with lightning protector element in place.

3.3.4.3 Voltage Standing Wave Ratio (VSWR). The VSWR of all low-loss coaxial cable assemblies in place shall not exceed 1.30 anywhere in the frequency range of 4.4 to 5.0 Gigahertz when tested in accordance with 4.6.4.3.

3.3.5 Cable Assemblies, Power and Audio Frequency. All power and audio frequency cable and cord assemblies shall be fabricated in accordance with the model unless otherwise specified and meet the continuity, dielectric, insulation resistance, and pull-out requirements of MIL-C-3885. (See 4.4)

3.4 Electrical Performance, Shelter Assemblage. The shelter assemblage shall operate error free on traffic and order wire channels when tested in accordance with 4.7.

3.5 Vehicular Transportation. The shelter assemblage shall be capable of being transported over cross-country terrain by military vehicle without sustaining any damage such as buckling, cracking, delamination, or other permanent deformation. (See 4.8)

3.6 Railroad Transportation. The shelter assemblage shall be capable of being loaded, blocked, and braced on a flatcar for shipment by rail and shall be subjected to and withstand the tests of 4.9 without sustaining any permanent damage such as buckling, cracking, delamination, or other permanent deformation.

3.7 Drop Test. The shelter assemblage shall be capable of withstanding drops of eighteen (18) inches onto concrete without sustaining any permanent damage such as buckling, cracking, delamination, or other permanent deformation. (See 4.10).

3.8 Air Tightness Test. The shelter facility shall show no evidence of air leakage through seams, joints, rivets, etc, or modified exterior surfaces when tested for tightness in accordance with 4.11.

3.9 Humidity. Hardware sample shall have no evidence of corrosion, rust, or deterioration after testing. (See 4.12)

3.10 Mounting Inserts. All mounting inserts shall be installed without the use of auxiliary materials such as adhesive and, when installed, each insert shall be capable of withstanding the tensile load and the torque load referenced in 4.13. There shall be no damage or permanent distortion to the shelter mounting surface or insert head and threads and the inserts shall not have rotated from their original position. Inserts and threads shall be free of paint.

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3.11 Cleaning. (See 3.22)

3.11.1 Parts. After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

3.11.2 Units. After assembly, the equipment shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware, and other foreign material.

3.12 Finish. The shelter facility shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 3.22 and 4.4)

3.13 Marking. (See 3.22 and 4.4)

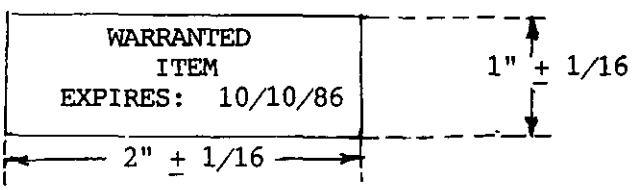
3.13.1 General. Marking shall conform to MIL-M-13231 and the drawings, unless otherwise specified.

3.13.2 Nameplates. Shelter facility and assemblage shall be provided with a nameplate in accordance with the drawings. Each nameplate shall be imprinted with a sequential serial number.

3.13.3 Aircraft Loading Data Plate. The shelter assemblage shall be provided with data plates in accordance with the drawings, imprinted with the dimensions which locate the center-of-gravity of the shelter facility and the assemblage in the three planes; the gross weight of the shelter facility and assemblage and the bearing pressure on the bottom of the skids. In the event of a design modification, these items will be changed as determined by the contractor using the first article of production.

3.13.4 Interior and Exterior Marking. All interior and exterior markings shall be as defined on the drawings, unless otherwise specified.

3.13.5 Warranty Markings. The contractor shall prepare and affix, to each warranted item furnished on contract, a warranty label as shown below. The information included on the label shall, as a minimum, identify the item as warranted materiel and provide the warranty expiration date. The marking(s) shall be in accordance with requirements of MIL-STD-130.



3.14 Riveting. Riveted joints shall be tight; the joined parts shall be undamaged; and the rivet heads shall be properly seated and tight against their bearing surfaces. Rivets shall conform to requirement 12 of MIL-STD-454. (See 3.22 and 4.4)

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3.15 Seal, Water Vapor. All joints of the framing for the video, fiber optic, power and antenna entrance boxes and air-conditioner ports shall be sealed to provide a barrier against the entrance of water vapor to the core material of the shelter panels and to the interior of the shelter. All joints and edges shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be coated by a one-part sealant Silastic 738, as made by Dow Corning, or equal using the manufacturers recommended primer. (See 3.22 and 4.16)

3.16 Securing of Parts. Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests. (See 3.22 and 4.16)

3.17 Soldering. (See 3.22 and 4.16)

3.17.1 Solder and Flux. Solder and flux shall be as defined on the drawings, unless otherwise specified.

3.17.2 Acid or Acid Salts. No acid or acid salts shall be used in preparation for or during soldering; however, exception permitted for preliminary tinning of electrical connections and for soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.17.3 Process. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign materials remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.18 Tropicalization of Material. Canvas, webbing, wood etc, shall be treated to be resistant to moisture and fungi as indicated on the drawings as follows:

3.18.1 Treating Material. Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.18.2 Toxicity. Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

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3.18.3 Flexibility. Treatment shall not affect the flexibility of treated materials to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.18.4 Statement of Treatment. The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.7)

3.19 Welding. Welding joints shall be as defined by the drawings unless otherwise specified and shall be such that grinding on the finished weld will be unnecessary except where grinding is specified on the drawings. Inert gas shielded arc welding (MIG) using helium, argon or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 CL.B and MIL-W-42505 CL.B as applicable. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except were spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4)

3.20 Wiring and Cabling. (See 3.22 and 4.16) Wiring and cabling shall be in accordance with the drawings unless otherwise specified and shall be neat and sturdy.

3.20.1 Slack. Wires and cables shall be as short as practical except that sufficient slack shall be provided:

a. To prevent undue stress on cable forms, wires, and connectors.

b. To permit removal from rack of operational equipments, having wiring or cables to connectors at back of equipment, so that cable may be connected or disconnected.

c. To facilitate field repair of broken or cut wires.

d. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.20.2 Protection. Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be sharply bent where they enter insulation materials.

3.20.3 Splicing. Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.20.4 Solderless Terminal Lug Connections. Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of MIL-T-7928. (See 4.4 and 4.14)

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3.20.5 Identification of Wiring. Wiring shall be identified by color coding and wire pair marking.

3.20.6 Cord Assemblies. Power cable and cord assemblies shall be fabricated in accordance with the drawings unless otherwise specified and meet the requirements and tests of MIL-C-3885. (See 4.4)

3.20.7 Testing. Power and signal wiring of the S-667/TRC-138A shall be subjected to and pass the tests described in 4.6.

3.21 Interchangeability. Like units assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.15) Individual items shall not be hand picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.22 Workmanship. (See 4.16) The equipment shall be manufactured and assembled in accordance with requirement 9 MIL-STD-454 and the applicable portions of the following paragraphs:

3.2 Construction
3.3 Electrical Installation
3.10 Mounting Inserts
3.11 Cleaning
3.12 Finish
3.13 Marking
3.14 Riveting
3.15 Seal, Water Vapor
3.16 Securing of Parts
3.17 Soldering
3.18 Tropicalization
3.19 Welding
3.20 Wiring and Cabling

3.23 EMI/EMC. (See 4.19 and 6.2.1) The AN/TRC-138A shall meet the following emission and susceptibility requirements of MIL-STD-461:

CEO2	CSO2A	REO2	RSO2
CEO3	CSO6	REO2.1	RSO3
CEO4A			
CEO5			

Notes:

a. REO2.1 shall be be applicable over the frequency range of 14 KHz to 1000 MHz.

b. RSO3 shall be applicable over the frequency range of 10 KHz to 1000 MHz.

c. Conducted emission and susceptibility requirements are applicable to 60 Hz and 400 Hz power lines.

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3.24 Systems Safety Engineering.

3.24.1 Personnel Hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be visual inspection. (See 4.16)

3.24.2 Edge Rounding Exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1mm), and exposed corners to a minimum of 0.05 inch (1.3mm).

3.24.3 Radioactive Material. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors, and lenses).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for Compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of Inspection. The inspections shall be classified as follows:

a. First Article inspection (does not include packaging). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspections of equipment before packaging. (See 4.5)

d. Packaging inspection. (See 4.17)

4.3 First Article. Unless otherwise specified in the contract or purchase order the first article inspection shall be performed by the contractor. (See 3.1)

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4.3.1 First Article Units. The contractor shall furnish one (1) first article unit of the complete AN/TRC-138A.

4.3.2 First Article Inspection. The first article inspection shall consist of the inspections specified in Table 1 and shall be performed in the order specified therein.

4.3.3 First Article Data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

TABLE I - FIRST ARTICLE INSPECTION

Inspection	Req't Para	Test Para
1. Inspection covered by subsidiary documents	---	4.4
2. Interchangeability	3.21	4.15
3. Mounting insert	3.10	4.13
4. Solderless-terminal-lug	3.20.4	4.14
5. Humidity	3.9	4.12
6. Visual and Mechanical	3.21, 3.22	4.16
7. System Safety Engineering	3.24	4.20
8. Electrical and RF installation	3.3	4.6
9. Electrical performance, shelter assemblage	3.4	4.7
10. Vehicular transportation	3.5, 3.16	4.8
11. Railroad transportation	3.6, 3.16	4.9
12. Drop tests	3.7, 3.16	4.10
13. Air-tightness	3.8	4.11
14. Repeat 8-Electrical and RF installation	3.3	4.6
15. Repeat 9-Electrical performance, shelter assemblage	3.4	4.7
16. Electromagnetic interference/ electromagnetic compatibility (EMI/EMC)	3.23	4.19

4.4 Inspection Covered by Subsidiary Documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order:

Item	Where Required
Cable and cord assemblies	3.3.5 & 3.20.6
Finish	3.12
Marking	3.13
Riveting	3.14
Welding	3.19
Solderless terminal lug connections	3.20.4

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4.5 Quality Conformance Inspection of Equipment Before Packaging. The contractor shall perform the inspection specified in paragraph 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A Inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQLs indicated in Table II, or an approved alternate Quality Audit (QA) procedure as set forth in 4.5.1.2.

4.5.1.1 Order of Inspection within Group A. Group A inspection shall be performed in the order listed in Table II.

TABLE II - GROUP A INSPECTION

Inspection	Req't Para	Test Para	AQL % Major	AQL % Minor
Visual and Mechanical	3.22	4.16	1.0	4.0
Electrical Installation	3.3	4.6	1.0	*

* All defects are major.

4.5.1.2 Alternate Group A Audit Inspection Procedures. The contractor may elect to use a continuous sampling and inspection plan instead of the lot inspection plans from MIL-STD-105. If this alternate is selected the procedure of implementation must be in accordance with MIL-STD-1235, "Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes." Sampling plans for AQLs listed in Table II Group A inspection shall be selected from Tables VIII, IX and X of MIL-STD-1235 which are for f=1/2. The number of units in the Production Interval shall be the entire production quantity unless otherwise specified in the contract. Before starting production the contractor shall notify the contracting officer in writing which method he intends to use. The method may not be changed during production without written approval of the Government.

4.5.2 Group B Inspection. No Group B inspection is required.

4.5.3 Group C Inspection. Group C inspection shall be performed on units that have passed Group A inspection. The inspection shall consist of the inspections specified in Table III. Samples shall be selected in accordance with 4.5.3.1.

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TABLE III Group C Inspection

Inspection	Req't Para	Test Para
Subgroup I		
Solderless-terminal-lug connection	3.20.4	4.14
Mounting insert (1000 lb., etc.)	3.10	4.13
Air-tightness	3.8	4.11
Interchangeability	3.21	4.15
System Safety Engineering	3.24	4.20
Subgroup II		
Humidity	3.9	4.12

4.5.3.1 Sampling for Group C Inspection.

4.5.3.1.1 Subgroup I. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every month. For the solderless-terminal test, sampling shall be as defined in 4.5.3.1.3.

4.5.3.1.2 Subgroup II. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fourth month. A unit shall consist of the specimens identified in 4.12.

4.5.3.1.3 Sampling for Inspection of Solderless Terminal Lug Connections. A sample for the test shall consist of five (5) specimens of each combination of wire and terminal produced each week. A specimen is a one (1) foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.2 Order of Inspection Within Group C. Group C inspection shall be performed in an order which is satisfactory to the Government.

4.5.3.3 Group C Failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of Conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on contract provided that they are resubjected to and pass Group A inspection after repair of all visible or known damage.

4.6 Electrical Installation Tests. The shelter facility shall be subjected to the tests of 4.6.1 through 4.6.4 below. Before applying power to the shelter facility, an electrical ground connection at the power entrance box shall be made. Power to be supplied shall be 120/208 VAC, 3 phase, 60 Hertz.

4.6.1 Power Systems and Components Installation Tests. Refer to the power schematic.

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a. Remove or disconnect the following electrical loads:

(1) All fluorescent, incandescent and neon lamps.
(2) All fluorescent lamp starters.
(3) Heaters.

b. Remove power distribution box cover panel and disconnect conductors from the voltmeter terminals. Position or isolate the disconnected leads so as not to short each other or facility ground.

c. Place meter switch on power distribution panel in the "OFF" position.

4.6.1.1 Continuity of Conductors. Continuity resistance tests which meet the requirements of paragraph 3.3.1.1 shall be performed on each power line circuit in accordance with the following paragraphs. A calibrated low resistance ohmmeter or equivalent shall be used in performing the test.

a. Place the following controls in the ON position:

(1) All circuit breakers.

(2) Fluorescent lamp switches.

(3) Incandescent lamp switches.

(4) Blackout bypass switch (to bypass position).

(5) Ground-fault receptacle (press "Reset" button to insure that ground fault receptacle is "ON").

b. Measure continuity resistance from pin A of power connector at the power entrance box to each of the following:

(1) Brass (hot) contacts of all convenience receptacles.

(2) Black wire termination at each fluorescent and incandescent lamp.

(3) Black wire termination at both ECU power cables.

(4) Brass (hot) contact of power indicator light socket.

(5) Black wire termination of ground fault receptacle in the power entrance box.

c. Measure continuity resistance from pin B of power connector at the power entrance box to each of the following.

(1) Brass (hot) contact of heater-1 receptacle.

(2) Red wire termination at both ECU power cables.

(3) Red wire termination of each equipment power cable.

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d. Measure continuity resistance from pin C of power connector at the power entrance box to each of the following:

(1) Brass (hot) contact of heater-2 receptacle.

(2) Blue wire termination at both ECU power cables.

(3) Blue wire termination of each equipment power cable.

e. Place all controls listed in step A in the OFF position except main circuit breaker.

f. Measure continuity resistance from pin N of power connector at the power entrance box to the following:

(1) White wire termination at ground fault receptacle in power entrance box.

(2) Silver contact of power indicator light, heater receptacles and convenience receptacles.

(3) White wire termination at each fluorescent and incandescent lamp.

(4) White wire termination at each equipment power cable.

g. Place main circuit breaker in the OFF position.

h. Measure continuity resistance from pin G (ground) of power connector at the power entrance box to each of the following:

(1) Ground terminal at each power receptacle and power cable.

(2) Ground terminal located at power entrance box, power distribution box, video entrance box, equipment racks and raceways.

4.6.1.2 Dielectric Breakdown and Insulation Resistance. The dielectric breakdown and insulation resistance of all power wires, cables, and components shall meet the requirements of paragraph 3.3.1.2. A nondestructive DC dielectric tester and a megohmmeter or equivalent shall be used in performing the tests. Test procedures shall be as follows:

CAUTION: Take preventive measures that will preclude personnel from coming into contact with conductors under test during performance of ALL dielectric and insulation resistance tests. Discharge each conductor to ground following each test in order to eliminate the danger of electric shock.

a. Place all circuit breakers and switches in the ON position except the fluorescent lamp switch. Place blackout/bypass switch to the bypass position.

b. Disconnect ground fault receptacle.

c. Disconnect at the meter switch all lines to the phase indicator circuitry. Isolate disconnected lines so as not to short to each other or to shelter facility ground.

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d. Test between pin A of the power input connector and all other pins of the power input connector strapped to shelter facility ground.

e. Place fluorescent lamp switches in the ON position.

f. Test between pin B of the power input connector and all other pins of the power input connector strapped to shelter facility ground.

g. Test between pin C of the power input connector and all other pins of the power input connector strapped to shelter facility ground.

h. Test between pin N of the power input connector and all other pins of the power input connector strapped to shelter facility ground.

4.6.2 Power distribution and operational function tests. Power distribution and operation function tests shall be performed on each power line in accordance with the following paragraphs.

4.6.2.1 Test Preparation.

a. Replace or reconnect all electrical loads, wires, and devices which were removed or disconnected by 4.6.1.a, 4.6.1.b and 4.6.1.2.b.

b. Close power distribution box panel cover.

c. Place all circuit breakers and switches in the OFF position.

d. Place blackout by-pass switch in the "ON" or bypass position.

e. Place a 115 volt AC test lamp with grounding plug into all convenience, heater, and intercom receptacles.

4.6.2.2 Test for 115 volt AC prime power.

a. Apply 120/208 VAC, 60 Hertz, 3 phase prime power to the shelter facility at the power connector. The power indicator lamp shall indicate power to the shelter.

b. Measure line voltage by connecting a calibrated AC voltmeter between terminal N and each of the Terminals A, B, and C on the power entrance panel.

c. Read voltage indicated on shelter facility voltmeter. The voltage reading shall be within $\pm$ 2% of the voltage measured by step B above.

d. Place main circuit breaker in the ON position.

4.6.2.3 Tests of Circuit Breakers and Switches.

a. Lights circuit breaker (CB 2).

(1) Place circuit breaker CB 2 in the ON position. Neon lamp above CB 2 shall light.

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(2) Place blackout bypass switch and fluorescent lamp switches in the ON position. All fluorescent lamps shall light.

(3) Place incandescent lamp switch in the ON position. All incandescent lamps shall light.

(4) Place CB 2 in the OFF position. All lamps mentioned in steps (1) thru (3) shall go out.

(5) Replace CB 2 in the ON position to restore light.

(6) With the shelter door open, place blackout bypass in the blackout position. All fluorescent and incandescent lamps shall go out.

(7) Close shelter door. All fluorescent and incandescent lamp shall light.

(8) Place fluorescent lamp switches in the OFF position. All fluorescent lamps shall go out.

(9) Place incandescent lamp switch in the OFF position. All incandescent lamps shall go out.

(10) Open door and restore lights by placing blackout bypass switch, fluorescent light switch, and incandescent light switch in the ON position.

b. Heater No. 1 circuit breaker (CB 3).

(1) Place circuit breaker CB 3 in the ON position. The neon lamp above CB 3 and test lamps in heater No. 1 receptacles shall light.

(2) Place circuit breakers CB 3 in the OFF position. The lamp mentioned in step (1) shall go out.

c. Heater No. 2 circuit breaker (CB 4).

(1) Place circuit breaker CB 4 in the ON position. The neon lamps above CB 4 and test lamp in heater No. 2 receptacle shall light.

(2) Place circuit breaker CB 4 in the OFF position. The lamp mentioned in step (1) shall go out.

d. Intercom and convenience circuit breaker (CB 5).

(1) Place circuit breaker CB 5 in the ON position. The neon lamp above the circuit breaker CB 5 and test lamps in the intercom and convenience receptacles shall light.

(2) Place CB 5 in the OFF position. The lamps mentioned in step (1) shall go out.

e. Equipment circuit breakers (CB 6 and CB 9).

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(1) Place circuit breaker CB 6 and CB 9 in the ON position. The neon lamps above CB 6 and CB 9 shall light.

(2) Using an AC voltmeter, measure the voltage between hot and neutral terminal (refer to power schematic) of each equipment power cable. The voltage measured shall be 115 ± 10%.

(3) Place circuit breakers CB 6 and CB 9 in the OFF position. The neon lamps above CB 6 and CB 9 shall go out.

(4) Repeat step (2) above. No voltage shall exist between the terminals indicated in the above step.

f. ECU-1 circuit breaker (CB 7).

(1) Place circuit breaker CB 7 in the ON position. The neon lamp above shall light.

(2) Using an AC voltmeter, measure the voltage between terminal D and each of terminals A, B, and C of the ECU-1 power cable. The voltage measured shall be 115 volts ± 10%.

(3) Place circuit breaker CB 7 in the OFF position. The neon lamp shall go out.

(4) Repeat step (2) above. No voltage shall exist between the terminals indicated in the above step.

g. ECU-2 circuit breaker (CB 8).

(1) Place circuit breaker CB 8 in the ON position. The neon lamp above CB 8 shall light.

(2) Using an AC voltmeter, measure the voltage between terminal D and each of terminals A, B, and C of the ECU-2 power cable. The voltage measured shall be 115 volts ± 10%.

(3) Place circuit breaker CB 8 in the OFF position. The neon lamp shall go out.

(4) Repeat step (2) above. No voltage shall exist between the terminals indicated in the above step.

h. External convenience circuit breaker (CB 10).

(1) Place circuit breaker CB 10 in the ON position.

(2) Press "RESET" on ground-fault receptacle to ensure it is on. Test lamps in the ground fault receptacle shall light.

(3) Press "TEST" on ground-fault receptacle and test lamps shall go out.

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(4) Press "RESET" on ground-fault receptacle. Test lamps shall light.

(5) Place circuit breaker CB 10 in the OFF position. The test lamps shall go out.

i. Main circuit breaker.

(1) Place all circuit breakers and switches in the ON position. All lamps shall light.

(2) Place main circuit breaker in the OFF position. All lamps shall go out.

(3) Place all circuit breakers and switches in the OFF position.

4.6.3 Signal System Wiring and Components Installation Tests. Refer to signal schematic.

4.6.3.1 Continuity Tests. The point-to-point continuity test of all signal lines shall meet the requirements of paragraph 3.3.3.1 as described in the following paragraphs. A low resistance continuity tester or equivalent shall be used to test continuity resistance.

a. Between each analog, digital voice orderwire, data orderwire, signal connector at the patch panel and its associated termination.

b. Between phone/intercom binding posts and their associated terminations at the phone/intercom location.

4.6.3.2 Dielectric Breakdown and Insulation Resistance Tests. The dielectric breakdown and insulation resistance of all signal lines, cables, and components shall meet the requirements of paragraph 3.3.3.2. A nondestructive tester shall be used in performing the tests. Test points are as follows: (See caution note under 4.6.1.2)

a. Each analog, digital voice orderwire, data orderwire and signal conductor at the patch panel to all other conductors of the same connector connected to ground.

b. Each phone/intercom conductor to ground.

4.6.4 Video System Wiring and Components Installation Tests.

4.6.4.1 Continuity Tests. The point-to-point continuity test (inner conductor to inner conductor and shield to shield) of all video circuits shall satisfy the requirements of 3.3.4.1. The test points are as follows:

a. Each video entrance box connector to its respective termination.

b. Each video connector at the patch panel to its respective equipment and termination.

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4.6.4.2 Dielectric and Insulation Resistance Tests. Dielectric and insulation resistance of all video lines, cables, and components shall meet the requirements of 3.3.4.2. A nondestructive tester shall be used in performing the tests. Test points are as follows: (See caution note under 4.6.1.2)

a. Each video entrance box connector and ground with the lightning protectors removed. Requirements of 3.3.4.2a shall be met.

b. Step a above with lightning protectors in place. Requirements of 3.3.4.2b shall be met.

c. Each remaining video patch panel connector to ground. Requirements of 3.3.4.2b shall be met.

4.6.4.3 Voltage Standing Wave Ratio (VSWR). VSWR which meets the requirements of 3.3.4.3 shall be performed on each low-loss transmission line in accordance with the following paragraphs. A system such as shown in Figure 5 for frequency range 4.4 to 5.0 Gigahertz shall be used.

a. Make necessary focus, intensity, and DC balance adjustments to the oscilloscope in accordance with the manufacturer's guidelines. Set oscilloscope to accept DC inputs. Set mode to LOG, Sensitivity to 5 dB/cm, and position vertically the trace at the bottom graticule.

b. Set sweep oscillator for automatic sweep between 4.4 and 5.0 GHz.

c. Connect the equipment for calibration purposes as shown in Figure 5.

d. Calibrate oscilloscope for direct return loss (dB) readout in accordance with the manufacturer's procedures.

e. With recorder turned OFF, connect the X and Y inputs to the oscilloscope's HORIZ and VERT outputs. Set sweep oscillator to manual sweep and manual sweep contol fully counterclockwise.

f. Set oscilloscope mode to CAL(dB)-0 and sensitivity to 10 dB/cm.

g. Set X-Y recorder vertical range to 100 mv/inch; and variable horizontal range to obtain full scale deflection.

h. Turn on the X-Y recorder and position the pen over the desired 0 dB (vertical) and start frequency (4.4 GHz) reference point on the graph paper using the recorder's zero controls.

i. Set manual sweep control of the sweep oscillator to fully clockwise and adjust oscilloscope horizontal amplitude to position pen to desired stop frequency (5.0 GHz reference). Check and re-adjust zero at start frequency reference if necessary. The recorder horizontal span is now calibrated.

j. Set oscilloscope mode to -10, -20, and -30 dB successively and adjust oscilloscope recorder output vertical amplitude so pen moves desired number of vertical divisions on the graph paper. Check and re-adjust vertical zero if necessary.

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k. Set oscilloscope mode to LOG. The recorder vertical span is now calibrated in dB/division corresponding to the oscilloscope sensitivity.

l. Insert system 1 low-loss cable as shown in figure 5.

m. Read return (dB) directly from the calibrated display.

n. Set the trigger control of the sweep oscillator to AUTO. Set sweep time for about 20-30 seconds.

o. Obtain an X-Y plot of the return loss by pushing the time button on the sweep oscillator.

p. Calculate VSWR on the plot at the worst condition(s) by the use of following formula. The requirements of 3.3.4.3 shall be met.

$$\log_{10} \rho = \frac{-\text{Return loss (dB)}}{20}$$

$$\text{VSWR} = \frac{(1 + \rho)}{(1 - \rho)}$$

Where ρ is the magnitude of the reflection coefficient.

q. Repeat VSWR test of steps a thru p above for system 2 and 3 respectively.

4.7 Electrical Performance, Shelter Assemblage. With all of the operational equipment installed in their respective rack locations and all cable assemblies connected thereto, perform the following interoperability tests.

Note: Testing shall be performed with COMSEC equipment or COMSEC equipment simulators installed unless otherwise indicated.

4.7.1 Equipment Interoperability, AN/GRC-222 Radio System.

a. Interconnect equipment as shown in Figure 1. Testing shall be at a group rate of 4608 kb/s.

b. Operate all equipment in use according to their respective technical manuals.

c. Verify analog voice orderwire (AVOW) operation.

d. Verify digital voice orderwire (DVOW) operation.

e. Verify data orderwire (DOW) operation.

f. Group traffic transmission shall meet the requirements of 3.4.

g. Verify display of faults from all equipment by alarm status indicator.

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4.7.2 Equipment Interoperability, AN/GRC-222 Radio System.

a. Interconnect equipment as shown in Figure 2. Testing shall be at a master group rate of 18.72 Mb/s.

b. Operate all equipment in use according to their respective technical manuals.

c. Verify analog voice orderwire (AVOW) operation.

d. Verify digital voice orderwire (DVOW) operation.

e. Verify data orderwire (DOW) operation.

f. Group traffic transmission shall meet the requirements of 3.4.

g. Verify display of faults from MGM and radio-1 by alarm status indicator.

4.7.3 Equipment Interoperability, Cable System.

a. Interconnect equipment as shown in Figure 3. Testing shall be at a master group rate of 18.72 Mb/s.

b. Operate all equipment in use according to their respective technical manuals.

c. Verify analog voice orderwire (AVOW) operation.

d. Verify digital voice orderwire (DVOW) operation.

e. Verify data orderwire (DOW) operation.

f. Group traffic transmission shall meet the requirements of 3.4.

g. Verify display of faults from HSCDMs by alarm status indicator.

4.7.4 Intercom and Radio. Verify operation of Intercommunications Station LS-147F/FI and Radio Set AN/VRC-46 by establishing voice communications between them and suitable external units.

4.8 Vehicular Transportation Test. The shelter assemblage shall be secured (and blocked to prevent shifting) to a Government furnished 5-ton, M-923, truck and transported over the specially prepared course described in paragraph 4.8.1 for ten (10) laps at each of the speeds 5 mph, 10 mph, 15 mph, and 18-20 mph. A lap is defined as traversing the course in one direction. After completion of the test, the shelter assemblage shall meet the requirements of paragraphs 3.5 and 3.16.

4.8.1 Vehicular Transportation Course. The course for the military truck transportation test shall have twelve (12) 4 x 8 inch timbers placed 25 feet apart on the 8 inch face and with the 4 inch face fully above ground. The top edges shall have 1 x 1 inch chamfers. The sixth and twelfth timbers shall be

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set 45 degrees to the direction to travel, and all other timbers shall be placed perpendicular to the direction of travel. The timbers shall be anchored down securely by any effective means. The surface on which the timbers are placed shall be hard and level.

4.9 Rail Transport, Impact Test. The shelter assemblage secured to a flatcar shall be impacted by a loaded gondola car. The test shall be performed on a straight and level stretch of track. The test flatcar with its timber blocking and shelter tiedown slings, shall be as specified in Figure 4 and shall meet the provisions of the test as specified in paragraph 4.9.1. The test flatcar (less the shelter) and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car shall meet the provisions of the test as specified in paragraph 4.9.2 and shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds traveling at 9 miles per hour (or equivalent energy conditions acceptable to the Government Technical Representative) shall be impacted against the stationary loaded test car coupled to the two empty flatcars with brakes off. The test shall consist of four (4) impacts. The first two impacts shall be with the door end of the shelter facing the loaded gondola car; the second two impacts shall be with the door end of the shelter facing the opposite direction from the loaded gondola car. The lashing, blocking, and wedges holding the shelter in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken during impact, the run shall be repeated. After each impact the shelter assemblage shall meet the requirements of paragraphs 3.6 and 3.16.

4.9.1 Test Flatcar. The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12 inch centers. The blocking along the door end and the opposite end of the shelter shall fit tightly against the corner castings along the full length of the castings except as necessary to clear the towing shackles, this shall be accomplished by means of power sawed notches. The blocking along the other two sides of the shelter shall fit tightly against the corner castings, along the full length of the castings. These four blocking members shall be notched or shimmed to contact the trim along the sides of the shelter. The 4 x 4 and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in Figure 4. All other blocking shall be firmly secured as shown in the figure. Lashing shall consist of four wire rope slings, each including 3/8 inch diameter, 6 x 19, flexible steel, performed wire rope (having a breaking strength of 10,6000 pounds minimum), turnbuckles, two hooks, with thimbles and fasteners as required; the breaking strength of the sling shall be 9000 pounds minimum. Before each impact, each sling shall be taut (600-800 pounds tension).

4.9.2 Loaded Gondola Car. The load in the loaded gondola car shall be one of the following:

a. A coarse size of either coal, crushed stone, gravel, concrete, or metal scrap (other than sheet metal, metal shavings, or other springy forms of metal).

b. Chunk, slab, block, or plate forms of the preceding acceptable materials, constrained to prevent shifting.

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4.10 Drop Tests.

4.10.1 Flat Drop. The shelter assemblage shall be lifted 18 inches from the ground and allowed to fall freely with the skids impacting onto a hard substantial concrete surface. The test shall be performed once. Upon completion of this drop, the assemblage shall meet the requirements of paragraphs 3.7 and 3.16.

4.10.2 Rotational Drop. The shelter assemblage shall have one skid (or the knuckles of the skids) resting on a nominal 4 inch high timber. The outer edge of the opposite skid (or the opposite knuckles of the skids) shall be raised 18 inches and the shelter shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four times, once for each bottom edge of the shelter. On completion of these drops, the shelter assemblage shall meet the requirements of paragraphs 3.7 and 3.16.

4.11 Air Tightness Test. The shelter facility, when subjected to the following test, shall meet the requirements of paragraph 3.8.

a. The air tightness shall be tested by introducing clean dry air into the closed shelter through an existing appropriate opening (i.e., drain fixture) to create an internal air pressure that exceeds the external by a minimum of 6 inches of water.

b. A manometer graduated 0.1 inches or less shall be utilized to measure this differential pressure.

c. The test apparatus shall include an air supply sufficient to maintain the required air pressure, an air flow regulator, and an input air pressure gage.

d. The shelter facility shall be tested for air tightness by painting the sealed seams, joints, rivets, etc, of the external modifications with Leak-Tac Formula No. 372 American Gas and Chemicals, Inc., Leak-Tac Division) or equal soap and water solution.

e. To minimize air loss, temporary seals, shims, and tape are permitted to seal the shelter door, blower cover, and air conditioner port opening not being tested.

4.12 Humidity Test. A hardware sample of five of each size and type (if the total quantity of any size or type is less than five, all shall be tested) of all fasteners shall be randomly selected for test. Hardware items which require welding shall have their finishes completed before testing. The hardware sample shall be subjected to the humidity test of MIL-STD-810, Method 507.1, Procedure II to determine conformance to the requirements of paragraph 3.9.

4.13 Mounting Insert Tests.

4.13.1 First Article and Group C Insert Test. Four of each size insert shall be randomly selected for test. Each insert shall be tested with an axial load followed by a torque load in accordance with Table IV. After test, the requirements of paragraph 3.10 shall be met. Failure of any insert shall be cause for rejection of the sample equipment. This test shall be conducted prior to installation of brackets, racks, etc.

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TABLE IV Mounting Insert Loads

First Article Test			Group C Test	
Insert	Axial (lbs)	Torque (in lbs)	Axial (lbs)	Torque (in lbs)
5/16-18	2000	100	1000	100
1/4-20	1300	60	650	60
#10-32	600	23	300	23

4.14 Testing of Solderless Terminal-Lug Connections. Connections made with solderless terminal-lug connectors shall be tested for conformance to paragraph 3.20.4.

4.15 Inspection for Interchangeability. The dimensions listed below shall be measured or gaged to determine compliance with the physical interchangeability requirements of paragraph 3.21. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting of brackets and mounting plates.

b. Size and mounting dimensions of racks and cabinets.

4.16 Visual and Mechanical Inspection. The shelter assemblage shall be examined for the defects listed in MIL-STD-252, the requirements of paragraph 3.22, and Table V. An inspection shall be performed to verify compliance with those portions of paragraph 3.24 (Systems Safety Engineering) which can be determined visually.

TABLE V Classification of Visual and Mechanical Defects

Classification	Defects
Major	1. Mounting plates missing or mislocated. 2. Aircraft loading data plate, instruction, or nameplate omitted, illegible or incorrect. 3. Welds - cracked or porous. Inspection of welded assemblies, such as equipment racks, entrance boxes, etc, should be made prior to final paint application. 4. Sealer improperly applied (holes, separations or lack of adhesion). 5. Mounting inserts missing or mislocated. 6. Burrs or sharp edges in wire duct not removed. 7. Equipments or bracketry loose or missing. 8. Doors, etc, inoperative. 9. Electrical receptacles, switches, jacks, connectors, wire ducts, or other electrical components loose, improperly located, or inoperative. 10. Locking or holding devices missing or inoperative. 11. Gaps between sections of wire duct exceed 1/16 inch.

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TABLE V Classification of Visual and Mechanical Defects (Cont'd)

Classification	Defects
	12. The air conditioner housing does not align properly with the mounting inserts of the shelter facility. 13. Grounds stud not properly sealed. 14. Bulkhead power and signal connector fasteners not sufficiently tight. 15. Electrical schematics missing.
Minor	1. Doors not easily operable. 2. Fastening devices difficult to operate. 3. Finish - abrasions or scratches. 4. Burrs and sharp edges on bracket not removed. 5. Screws loose. 6. Gaskets, not completely and securely adhered. 7. Electrical schematics not properly adhered to wall. 8. Cable assemblies not dressed properly.

4.17 Packaging Inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging.

b. Quality Conformance inspection of Packaging.

4.17.1 First Article Inspection Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.17.2 Quality Conformance Inspection of Packaging.

4.17.2.1 Materials Inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.17.2.2 Preservation Inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.17.2.3 Packing Inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table VI, "PACKING INSPECTION PROVISIONS". Log formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality level of 4.0 percent defective.

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TABLE VI PACKING INSPECTION PROVISIONS

No.	Characteristic	Method of Inspection
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

4.18 Rough Handling Test (Packaging). When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operation malfunction caused by rough handling:

Paragraph 4.6 Power and Signal Wiring
Paragraph 4.16 Visual and Mechanical

4.19 Electromagnetic Interference Test. The complete assemblage shall be tested to determine compliance with the requirements of paragraphs 3.23 and 6.2.1. Test methods and procedures shall be in accordance with MIL-STD-462, as implemented by a contractor-prepared, Government-approved, EMI Test Plan as part of the First Article Test Plan.

4.20 System Safety Engineering. Using SEL Form 1183, verify that the requirements of 3.24 are met.

5. PACKAGING.

5.1 Packaging Requirements. Packaging requirements shall be in accordance with MIL-S-55507.

6. NOTES.

6.1 Intended Use. The major operational equipments comprising the AN/TRC-138A are as follows:

a. 1 ea. Shelter, Electrical Equipment S-667/TRC-138A
b. 3 ea. Radio Set AN/GRC-222
c. 3 ea. Modem, Digital Data MD-1026(P)/G
d. 1 ea. Modem, Cable Driver MD-1024/G
e. 1 ea. Multiplexer, TD-1237(P)/G

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f. 1 ea. Encryption Device TSEC/KY-57
g. 1 ea. Control, Orderwire C-10717/TRC
h. 1 ea. Intercommunications Station LS-147F/FI
i. 1 ea. Digital Secure Voice Telephone TSEC/KY-68
j. 1 ea. Radio Set AN/VRC-46
k. 1 ea. Indicator, Alarm Status ID-2324(V)4/TRC
l. 1 ea. Azimuth - Elevation Control
m. 1 ea. Power Supply PP-2953C/U

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. First Article and Group C inspection of the shelter assemblage requires use of contractor fabricated dummy loads or the Government furnished live loads (operational equipment) as specified in the bid request and contract.

c. A list of Government furnished equipments required as part of the S-667/TRC-138A.

d. A list of equipments to be Government loaned for testing purposes.

e. Level A or B preservation and packing (see section 5).

f. When first article inspection rough handling tests are not required.

g. When first article packaging inspection test reports require Acquisition Activity approval prior to production unit packing.

6.2.1 Acquistion requirements. Acquistion documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation.

c. MIL-STD-461A, Notice 4 shall be used for EMI requirements in 3.23.

d. MIL-STD-462, Notice 3 shall be used for tests in 4.19.

6.3 Inspection. Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination. Examination consists of simple, generally nondestructive determinations of compliance, without use of special test equipment.

6.3.2 Testing. Testing consists of determination of compliance using technical means.

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision of the contracting officer on the adequacy of corrective action. (See 4.5.3.3).

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6.5 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: S-667/TRC-138A. The contractor shall apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Verification Inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Tropicalization of Material. All materials of construction are specified on the drawing and are fungus-inert. Substitution of any materials other than those specified must be approved by the contracting officer. The material will only be approved after the contractor tests the material for Fungus Resistance satisfactorily in accordance with requirement 508 of MIL-STD-810.

6.8 Level B Preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.9 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity, for recommended disposability methods.

6.10 Change from Previous Issue: Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue, due to the nature and extent of the changes.

The following major changes are included:

- Document revision has been changed to A.
- Sections 3, 4 and 6 have been revised to reflect the replacement of the AN/GRC-144(V)3 Radio by the AN/GRC-222 Radio.
- Deletion of high band operation (14.4 to 15.0 GHz).
- Replacement of waveguide by low-loss coaxial cable.
- The text and formatting of text is section 2 is modified.
- Requirement of Warranty Marking in section 3 has been added.
- The figures of radio system and VSWR measurement have been changed to reflect use of the AN/GRC-222 Radio.

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6.11 Subject Term (Keyword) Listing.

AN/TRC-138A
Group Traffic Transmission
Master Group Multiplexer
Radio Repeater Assemblage
Radio Set AN/GRC-222
Radio Park
S-667/TRC-138A
Shelter Facility
Shelter Assemblage
Shelter, Electrical Equipment
Voltage Standing Wave Ratio
Wiring Video System

Custodian:

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Preparing Activity

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Project 5820-A862

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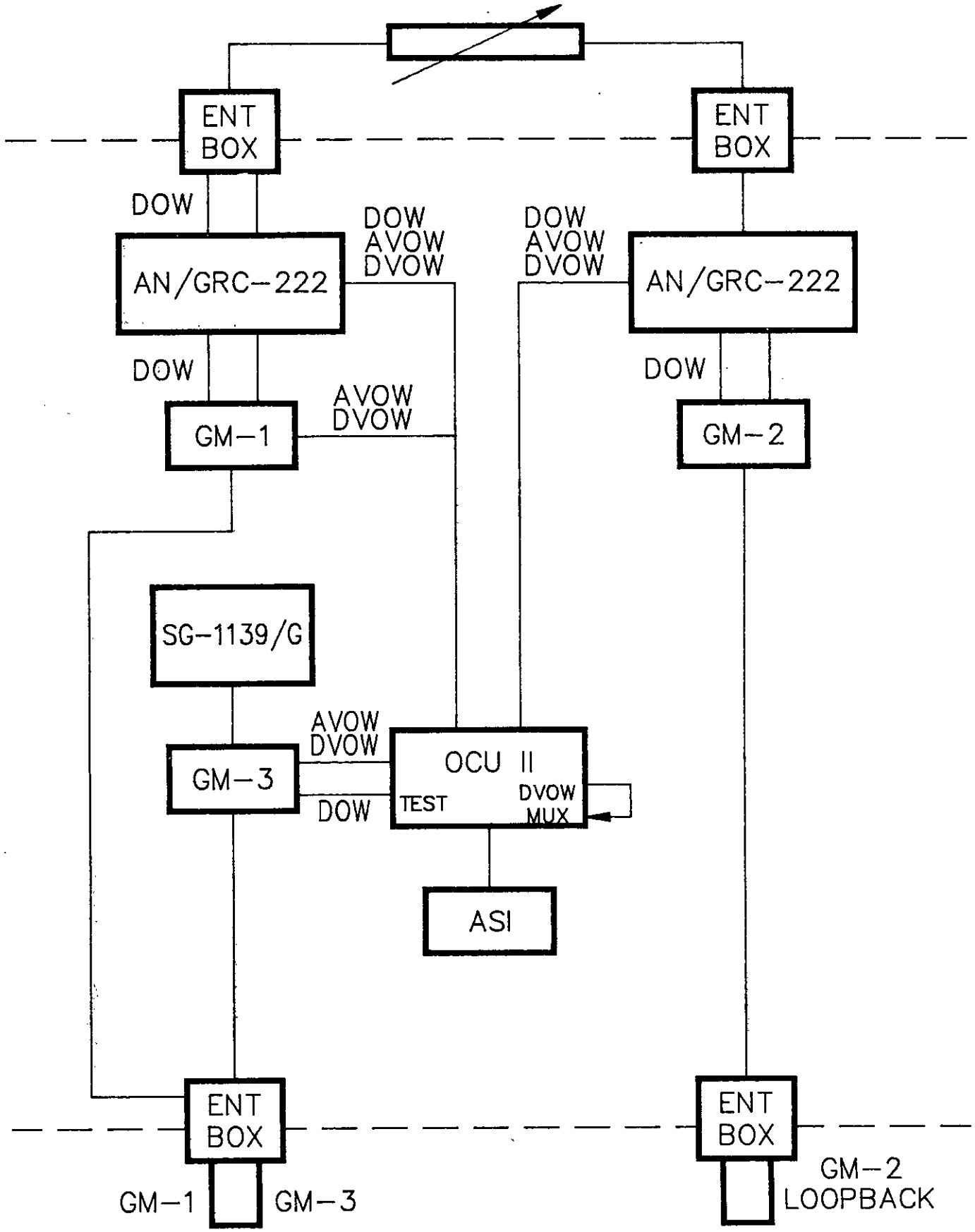


FIGURE 1. – RADIO AND GROUP MODEM SYSTEM

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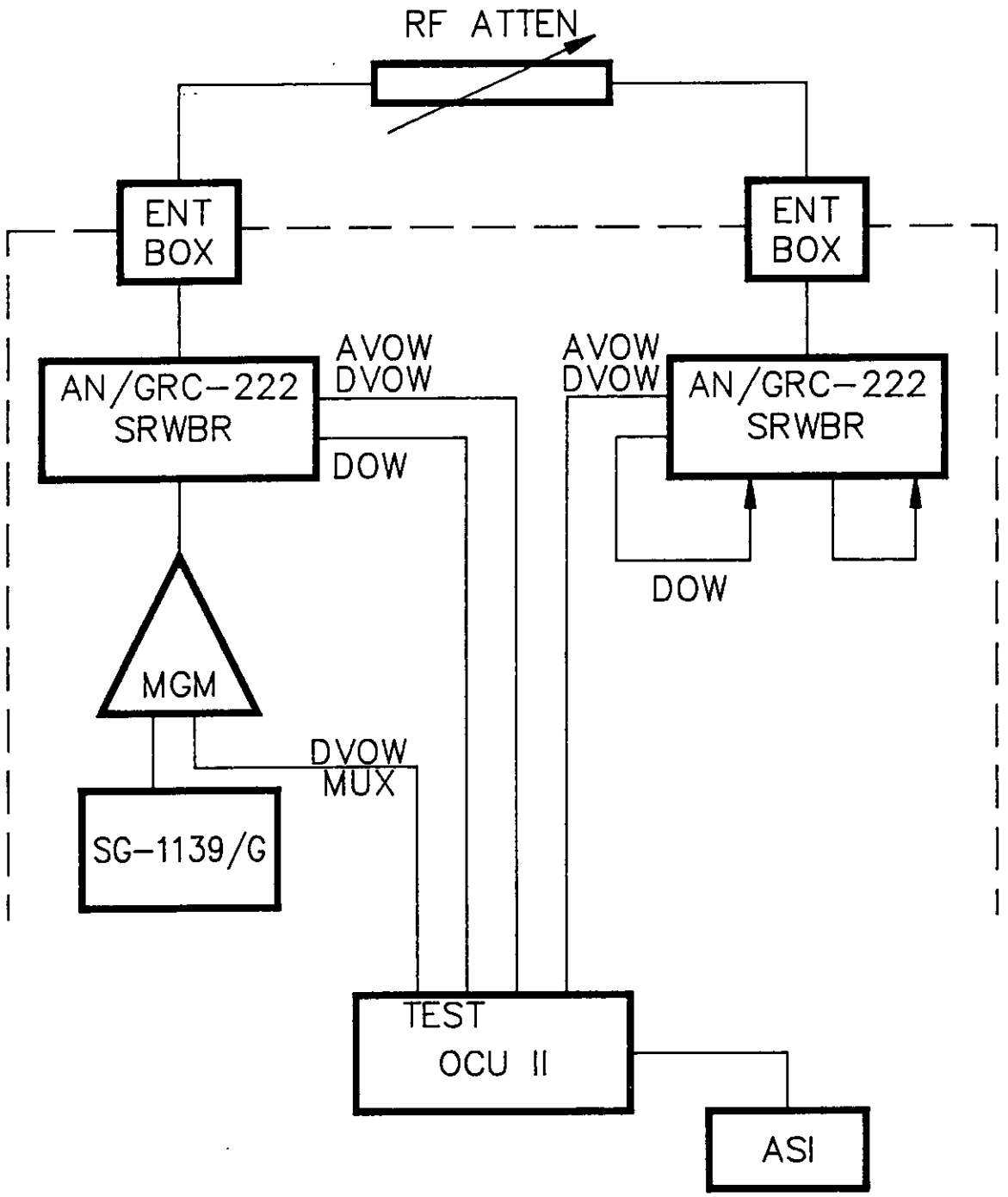


FIGURE 2 - RADIO AND MASTER GROUP MODEM SYSTEM

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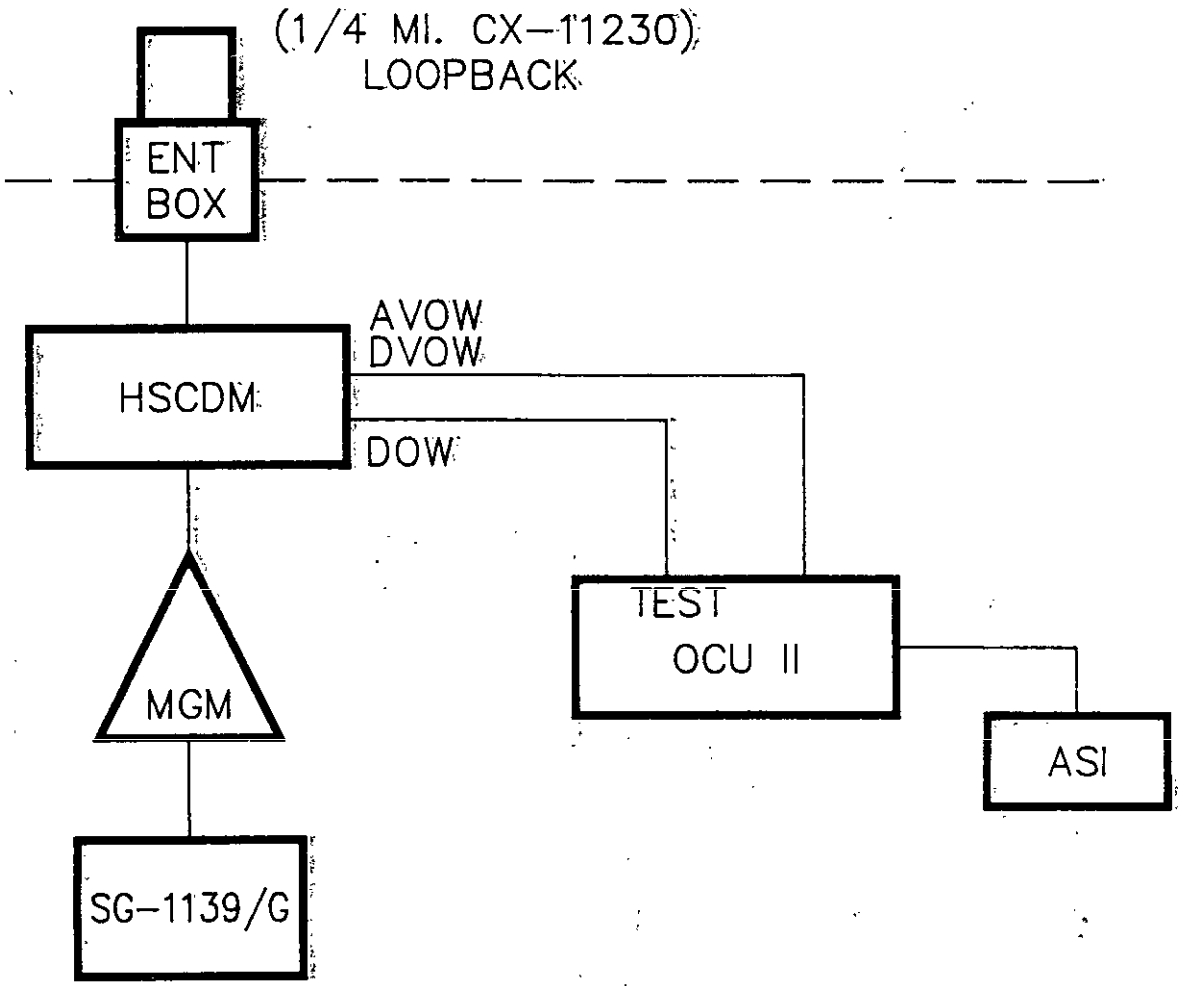
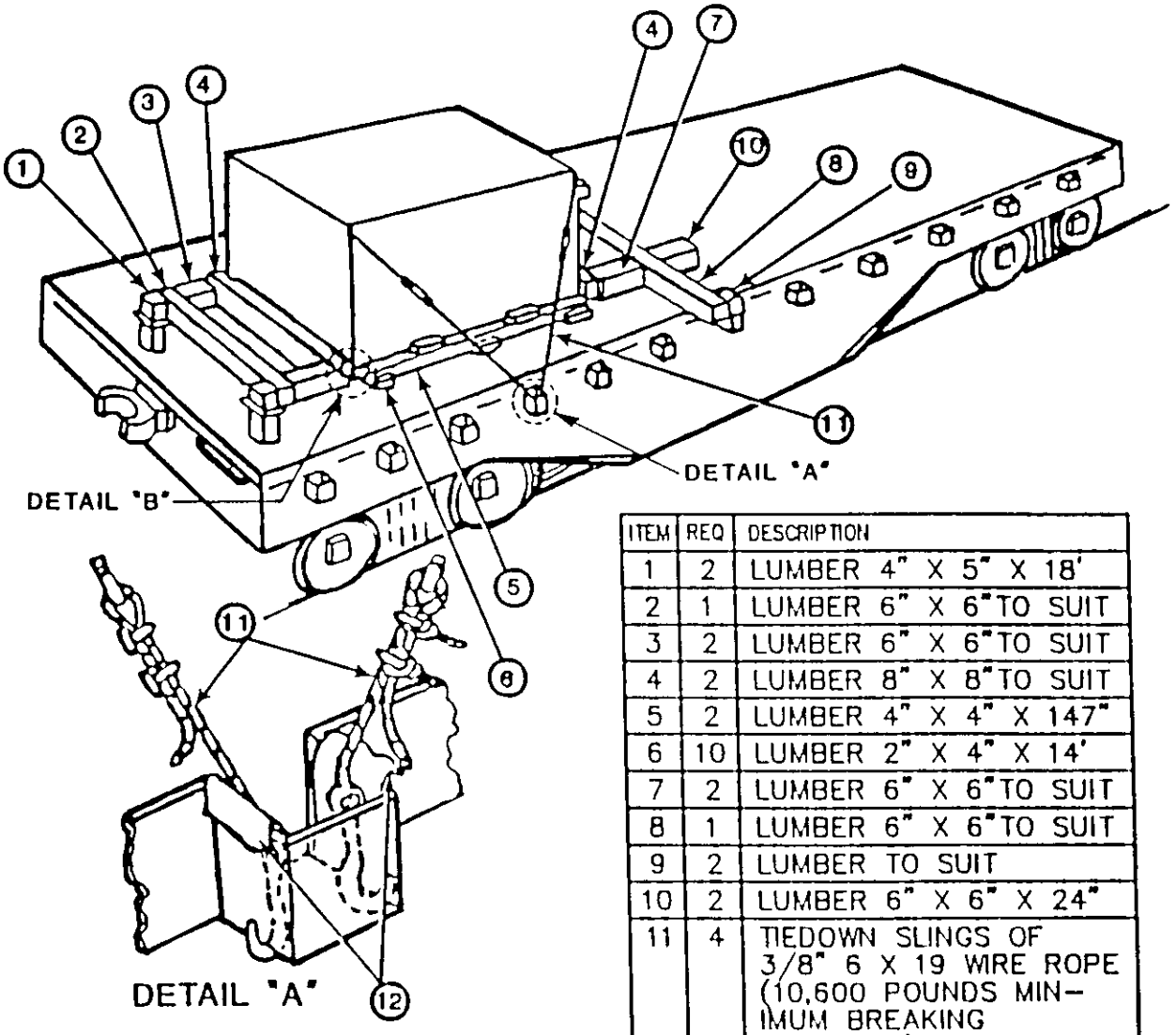
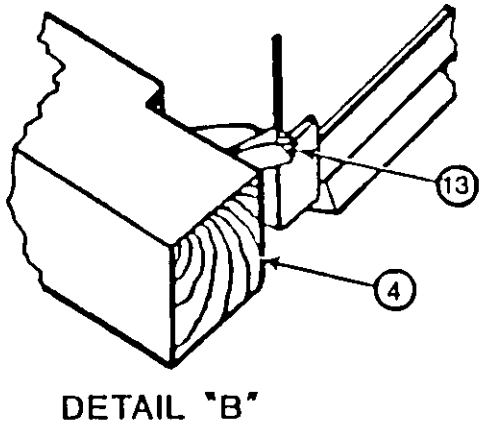


FIGURE 3 - CABLE SYSTEM

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ITEM	REQ	DESCRIPTION
1	2	LUMBER 4" X 5" X 18'
2	1	LUMBER 6" X 6"TO SUIT
3	2	LUMBER 6" X 6"TO SUIT
4	2	LUMBER 8" X 8"TO SUIT
5	2	LUMBER 4" X 4" X 147"
6	10	LUMBER 2" X 4" X 14'
7	2	LUMBER 6" X 6"TO SUIT
8	1	LUMBER 6" X 6"TO SUIT
9	2	LUMBER TO SUIT
10	2	LUMBER 6" X 6" X 24"
11	4	TIEDOWN SLINGS OF 3/8" 6 X 19 WIRE ROPE (10,600 POUNDS MIN-IMUM BREAKING STRENGTH). SEE 4.9.1
12	4	FLEX. METAL SHEETING
13	4	OAK WEDGES, 1" WIDE

NOTES: a. The blocking lumber shall be notched to accomodate towing eyes and bolts.

b. Wedge Item 13 between towing corner forging and Item 4 notch (Detail B).

c. The blocking lumber shall be rigidly fastened to the floor of the flatcar.

FIGURE 4 – RAIL TRANSPORT IMPACT TEST

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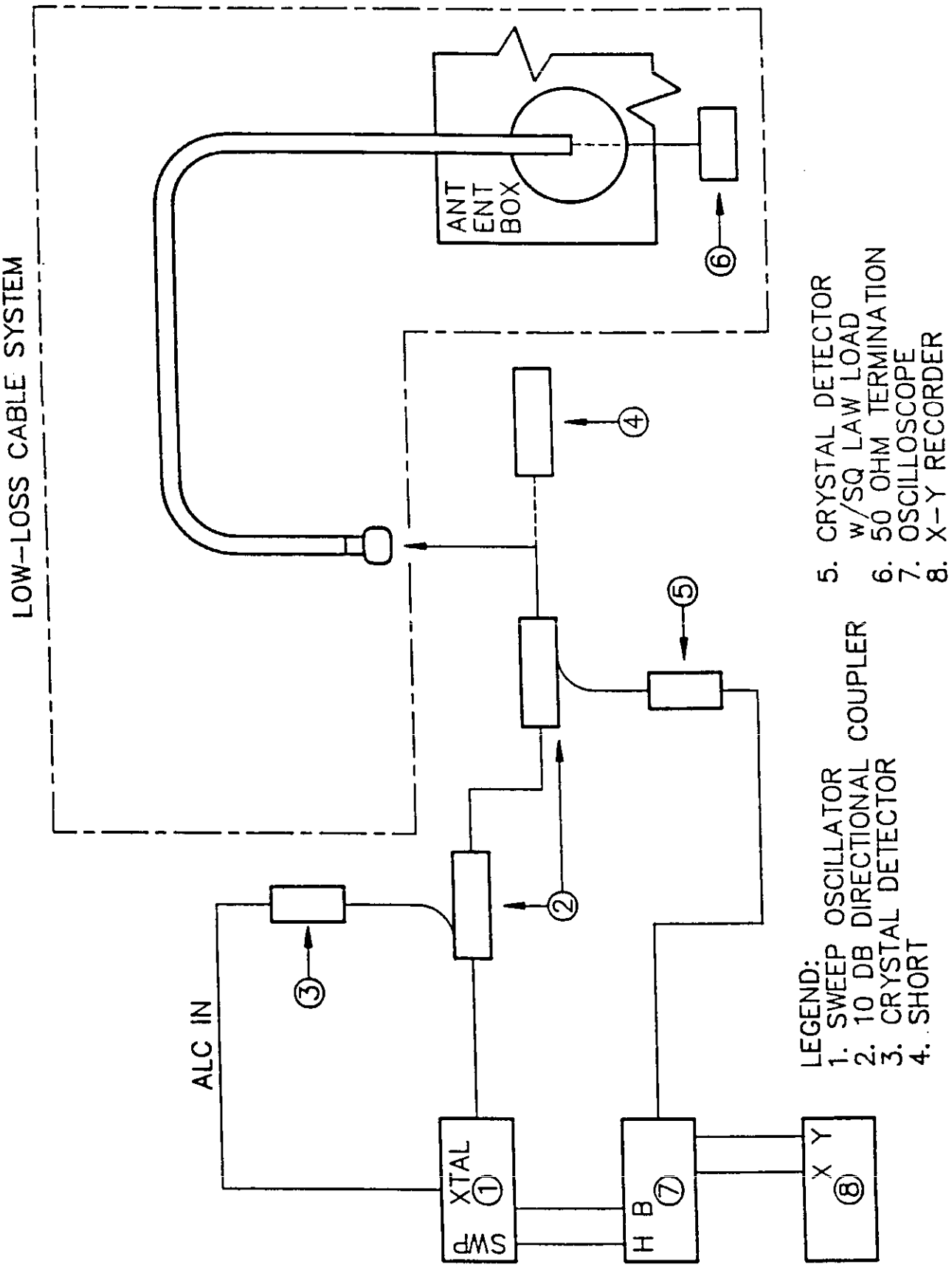


FIGURE 5 - TYPICAL VSWR TEST SETUP

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-49276A(CR)	Radio Repeater Set, AN/TRC-138A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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